



OCZ has launched their new “Z” series of power supplies which are 80 plus Gold certified – most efficient rating possible

New chip foundry

GlobalFoundries, the chip fabrication spin-off from AMD, gears up as it gets its first customer for 22nm; and it's not AMD, but ST Micro

tasks with ease – surfing, watching movies (including HD playback), basic gaming, office work. Single core processors are largely Jurassic and you shouldn't be buying one – more applications are slowly getting multithreaded and in order to run them optimally you need a multi-core CPU. CPUs in this price bracket are not meant for high-end gamers or rendering workstations. There are very few specifications we can recommend and we also suggest you stay away from the typical vendor jargon such as “you need at least a 2.0 GHz dual core” – processor clock speeds and the number of cores alone are not always a yardstick of their generic performance. Some CPUs are naturally better suited to some applications just as surely as most application developers design their software with optimisations for a particular CPU vendor.

AMD's Sempron 140 was the sole single core in the entire comparison and it's also the cheapest processor, at



Rs. 2,160. Shockingly, it bares its teeth quite successfully against Intel's two costlier dual core offerings – no doubt on account of its higher core clock and the higher FSB speed. Remember that Intel's basic dual cores run off an 800 MHz bus and these CPUs also miss

out on an integrated memory controller (IMC hereon). This is a double whammy for AMD, since we know that for any dual core *sans* IMC to perform, the FSB speed should be suitably high to prevent data bottlenecking in the FSB. Evidently, this is what

happens. Shame on Intel for not offering a faster FSB speed such as is present on its Core 2 Duo processors, but AMD isn't going to cry about this.

Just take a look at the ScienceMark 2.0 memory bandwidth scores – AMD gets an advantage owing to the minimal latency between the CPU and IMC which are on the same die. For Intel the memory controller resides on the Northbridge chip connected via its super slow 800 MHz FSB, which is why its scores dip, despite using fast DDR3 RAM. In fact, all four AMD processors are faster on the whole than their Intel counterparts. Three of the costlier AMD CPUs are faster at games – one application considered during the last four years to be Intel's forte. AMD's newer Athlon II processors are codenamed *Regor*, and are based on a 45-nm fabrication process and have relatively large caches of 256 kB (L1) and 2 MB (L2). These processors are the fastest in terms of clock speed in this group, the Athlon II X2 240 and X2 250 being 2.8 GHz and 3.0 GHz. The fact that the single core Sempron 140 can get a performance edge over Intel's dual-core offerings just demonstrates how important

